SMARTBRIDGE
Let's Bridge the Gap

Technology Stack

Project Design Phase-II (Architecture & Stack)

SKILLWALLET
A SmartBridge Product

Date	31 January 2026	**Team ID**	TEAM-FUEL-08
Project Name	Global Fuel Price & Taxation Trends	**Maximum Marks**	2 Marks

Define Your Technology Stack: Identify the technologies, platforms, databases, BI tools, and integrations used to build your Business Intelligence solution. A well-chosen technology stack ensures the solution is scalable, maintainable, secure, and optimized for global fuel price and tax reporting requirements.

Technology Component Selection & Purpose

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze fuel price datasets.	**Python, SQL, HTML/CSS**	Enables automated data cleaning, currency conversions (`petrol_usd_liter`, `diesel_usd_liter`), and custom layout styling for dashboard embeds.
2.	**Database / Data Warehouse**	Warehouse used to store and manage structured international economic data for reporting.	**GitHub Datasets / Snowflake / CSV**	Provides fast columnar querying, seamless integration with Tableau extracts, and structured storage for multi-country repository data (`global fuel/`).
3.	**Data Ingestion**	Used to load data from global source files into the analytical visualization environment.	**Tableau Data Connections**	Allows live and extract data connection pipelines directly to global fuel and crude oil benchmark datasets (`brent_crude_usd`).
4.	**Data Transformation & Business Logic**	Tools/logic used to clean, join, and calculate KPIs, tax ratios, and analytical measures.	**Tableau Prep / Desktop SQL**	Calculates key analytics measures like average petrol USD per liter, `tax_percentage` margins, `subsidy_level`, and `income_level` categorizations.
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, scatter plots, geospatial maps, and visual trends.	**Tableau Desktop**	Delivers rich interactive charts, cross-filtering, and dynamic trend lines correlating Brent Crude to domestic retail pricing.
6.	**Front-End / User Interface**	Interface through which users interact with dashboard filters, parameters, and visual insights.	**Tableau Public**	Provides a web-accessible interactive analytics platform for global stakeholders and executive policy reviews.